Solution:

Here, $\Delta ABC \sim \Delta ADE$ (given)

$\therefore$ Their corresponding sides are proportional (by RHS similarity).

$$\therefore \quad \frac{AC}{AE} = \frac{BC}{DE}$$

$$\Rightarrow \frac{12}{96} = \frac{5}{h}$$

$$\Rightarrow h = \frac{5 \times 96}{12} = 40 \text{ } feet$$

$\therefore$ The height of the tree is 40 *feet.*

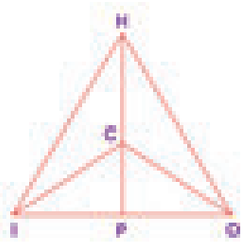
Activity

The teacher cuts many triangles that are similar or congruent from a card board (or) chart sheet. The students are asked to find which pair of triangles are similar or congruent based on the measures indicated in the triangles.

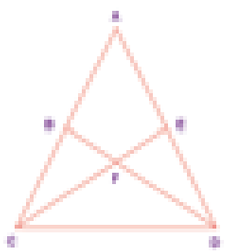
Exercise 5.1

1. Fill in the blanks with the correct term from the given list.
(in proportion, similar, corresponding, congruent, shape, area, equal)
   (i) Corresponding sides of similar triangles are _____.
   (ii) Similar triangles have the same _____ but not necessarily the same size.
   (iii) In any triangle _____ sides are opposite to equal angles.
   (iv) The symbol $\equiv$ is used to represent _____ triangles.
   (v) The symbol $\sim$ is used to represent _____ triangles.

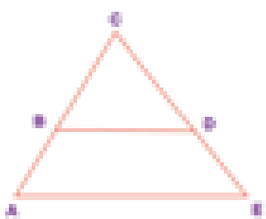
2. In the given figure, $\angle CIP \equiv \angle COP$ and $\angle HIP \equiv \angle HOP$. Prove that $IP \equiv OP$.



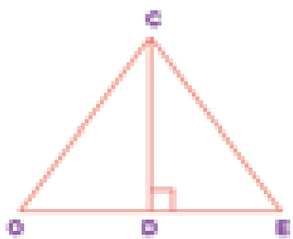
3. In the given figure, $AC \equiv AD$ and $\angle CBD \equiv \angle DEC$. Prove that $\Delta BCF \equiv \Delta EDF$.



4. In the given figure, ΔBCD is isosceles with base BD and $\angle BAE \equiv \angle DEA$. Prove that $AB \equiv ED$.



5. In the given figure, D is the midpoint of OE and $\angle CDE = 90°$. Prove that $\Delta ODC \equiv \Delta EDC$



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